

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

Bouke Klein Teeselink^{*1,2,3}

¹*Department of Political Economy, King's College London*

²*AI Objectives Institute*

³*King's Institute for Artificial Intelligence*

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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ket characteristics. We examine heterogeneity along two dimensions: compensation levels (splitting firms and occupations by median pre-treatment salary) and geographic concentration (comparing London-based versus non-London entities). These dimensions are highly policy-relevant, as compensation-level heterogeneity provides a first indication as to whether LLMs may compress or widen wage inequality, while geographic variation tests whether disruption concentrates in London’s knowledge economy or extends throughout the UK labor market.

4.3.1 Heterogeneity by Compensation Level

Table A2 examines how firm-level employment effects vary by average compensation, with high-compensation jobs serving as a proxy for more highly educated white collar work, and low-compensation jobs for relatively low-educated blue collar work. We split firms at the median pre-treatment compensation level. Panel A shows that high-compensation firms experience substantially larger employment reductions than their lower-paying counterparts. A one standard deviation increase in LLM exposure reduces total employment at high-compensation firms by 0.7 percent ($p = 0.009$), driven primarily by a 0.8 percent reduction in low-seniority positions ($p = 0.005$). In contrast, Panel B indicates that low-compensation firms show no significant employment effects, with point estimates near zero for both total employment and low-seniority positions.

This pattern extends to hiring intentions. Table A4 shows that a one standard deviation increase in LLM exposure reduces the probability of posting vacancies by 1.3 percentage points at high-compensation firms ($p < 0.001$), compared to a 0.7 percentage point reduction at low-compensation firms ($p = 0.001$). While both groups curtail hiring, the effect is approximately twice as large at high-compensation firms.

The occupation-level analysis in Table A5 provides further evidence that the employment effects are concentrated among the highly paid. In high-salary occupations, a one standard deviation increase in LLM exposure is associated with a 5.7 percent reduction in job postings ($p = 0.009$) and £782 salary declines ($p < 0.001$), while low-salary occupations show no significant changes in posting volumes and minimal salary effects.

These results suggest that LLM disruption is almost entirely driven by changes

in higher-wage segments of the labor market, contradicting historical automation patterns that predominantly affected routine middle-skill occupations. The concentration of employment losses and wage pressures in high-compensation firms and high-salary occupations implies that LLMs may have an equalizing effect on the income distribution, at least in the short run. This pattern aligns partially with the optimistic scenario outlined by Autor (2024) who argues that AI could help rebuild middle-class jobs by enabling workers with foundational expertise to perform higher-stakes decision-making tasks previously reserved for elite professionals. However, our findings suggest a more pessimistic mechanism: rather than elevating middle-skill workers to perform expert tasks, LLMs appear to be eroding demand for the expert tasks themselves. While this may compress wage inequality by reducing returns to education and expertise, it appears to achieve this through a leveling-down rather than the leveling-up process envisioned by Autor (2024).

4.3.2 Geographic Heterogeneity

Geographic concentration provides another lens for understanding differential treatment effects. Table A3 compares firms headquartered in London versus elsewhere in the UK. Surprisingly, London-based firms show minimal employment responses despite their concentration in LLM-exposed industries like finance and professional services. Total employment effects are statistically insignificant (0.1 percent per standard deviation), with similar null results for both seniority levels. In contrast, there is some evidence that firms headquartered outside London experience stronger employment declines from LLM exposure: a one standard deviation increase in LLM exposure reduces total employment by 0.4 percent ($p = 0.042$), with low-seniority positions declining by 0.5 percent ($p = 0.020$). That said, the firm-level hiring response shows the opposite pattern: a one standard deviation increase in LLM exposure reduces job postings by 1.6 percentage points for London-based firms versus 0.8 percentage points for firms outside London, despite their more modest employment stock adjustments.

At the occupation level, both London and non-London job markets experience significant disruption. A one standard deviation increase in LLM exposure is associated with a 3.4 percent reduction in job postings in London ($p = 0.047$) with salary declines of £563 ($p < 0.001$), while positions outside London experience posting